Phys 115B HW 9

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Question 1. *Consider the parity operator $\hat{\Pi}$ in three dimensions, such that $\hat{\Pi}\psi(\mathbf{r}) = \psi(-\mathbf{r})$.*

(a) Show that for states ψ expressed in polar coordinates, the action of the parity operator can be written as

$$\hat{\Pi}\psi(r, \theta, \phi) = \psi(r, \pi - \theta, \phi + \pi)$$

(b) Show that for hydrogenic energy eigenstates $\psi_{nlm}(r, \theta, \phi)$ we have

$$\hat{\Pi}\psi_{nlm} = (-1)^l \psi_{nlm}(r, \theta, \phi)$$

Proof. □

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Question 2. *We can classify operators depending on how they transform under parity,* $\hat{Q} \to \hat{\Pi}^\dagger \hat{Q} \hat{\Pi}$ *(see example 6.2 or Table 6.1 in the book can be useful).*

(a) *In one dimension, show that position and momentum are odd under parity.*

(b) *In three dimensions, operators that are invariant under parity and commute with* $\hat{L}$ *are called scalar operators,* $\hat{\Pi}^\dagger \hat{Q}_s \hat{\Pi} = \hat{Q}_s$. *Operators that flip sign under parity and commute with* $\hat{L}$ *are called pseudo-scalar operators,* $\hat{\Pi}^\dagger \hat{Q}_s \hat{\Pi} = -\hat{Q}_s$. *Show that scalar operators satisfy* $[\hat{\Pi}, \hat{Q}_s] = 0$, *while pseudoscalar operators satisfy*

$$\{\hat{\Pi}, \hat{Q}_s\} = 0$$

(where the curly brackets denote the anti-commutator).

(c) *Operators that flip sign under parity and whose commutators with* $\hat{L}$ *are like those of* $\hat{L}$ *are called vector operators, such that* $\hat{\Pi}^\dagger \hat{Q}_v \hat{\Pi} = -\hat{Q}_v$. *Operators that are invariant under parity and whose commutators with* $\hat{L}$ *are like those of* $\hat{L}$ *are called pseudovector or axial vector operators,* $\hat{\Pi}^\dagger \hat{Q}_a \hat{\Pi} = \hat{Q}_a$. *Show that vector operators satisfy*

$$\{\hat{\Pi}, \hat{Q}_v\} = 0$$

and pseudo-vectors satisfy

$$[\hat{\Pi}, \hat{Q}_a] = 0.$$

(d) *Give an example of a scalar operator, pseudoscalar operator, a vector operator and an axial or pseudo vector operator (feel free to Google or use the book).*

Proof. $\square$

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Question 3. *Consider a particle of mass m and charge q confined to a circular ring in the x-y plane, with radius r. The system will be driven by an electric field E_x in the x direction (which can stimulate photon induced transitions). It is natural to work in polar coordinates for this problem, with $x = r\cos\theta$.*

(a) What are the first five energies of the system with $E_x = 0$?

(b) What are the first five eigenfunctions $\psi(\theta)$? Please normalize them, and pick your basis such that the eigenfunctions are also eigenfunctions of L_z.

(c) Explicitly write out a 3×3 matrix for the Hamiltonian

$$H' = E_x q x = E q r \cos\theta$$

This matrix will use the three lowest states above. Extrapolate this result to write a concise selection rule that tells what transitions are allowed for any state.

(d) What frequencies are absorbed by a "gas" of such rings if they are cold enough to be initially in the ground state?

(e) What additional frequencies are absorbed by rings initially in the first excited state?

Proof. □